



Short communication

Cannabis expectancies mediate the association between social media use and cannabis experimentation in early adolescents: A prospective cohort study

Jason M. Nagata^{a,*}, Andrew Caffrey^a, Nathan D. Nguyen^a, Sahana Nayak^a, Isaac Frimpong^a, Christiane K. Helmer^a, Colbey Ricklefs^a, Abubakr A. Al-Shoaibi^a, Alexander Testa^b, Claire D. Brindis^{a,c}, Glenn-Milo Santos^d, Fiona C. Baker^e

^a Department of Pediatrics, University of California, San Francisco, 550 16th Street, Box 0503, San Francisco, CA 94143, USA

^b Department of Management, Policy and Community Health, University of Texas Health Science Center at Houston, 7000 Fannin, Suite 1706, Houston, TX 77030, USA

^c Philip R. Lee Institute for Health Policy Studies, University of California, San Francisco, 490 Illinois St, San Francisco, CA 94158, USA

^d Department of Community Health Systems, School of Nursing, University of California, San Francisco, 2 Koret Way, San Francisco, CA 94143, USA

^e Center for Health Sciences, SRI International, Menlo Park, 333 Ravenswood Ave, Menlo Park, CA 94025, USA

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ABSTRACT

Background: Social media exposure may influence early cannabis use behaviors in adolescents, potentially increasing the risk of future problematic use. Minimal prior research has investigated the role of cannabis expectancies (i.e., beliefs about the anticipated positive or negative effects of cannabis) and their role in mediating cannabis use initiation in early adolescence.

Objective: To examine the prospective association between social media use and cannabis experimentation in a diverse U.S. sample of early adolescents, and to determine the extent to which cannabis expectancies mediate this relationship.

Methods: We utilized longitudinal data from the Adolescent Brain Cognitive Development Study ($N = 7691$, mean age 12.0 at Year 2 follow-up, 2018–2020) to estimate associations between daily social media time (Year 2) and cannabis use outcomes (Year 4) using sequential modeling approaches, adjusting for confounders including friends' cannabis use, and testing cannabis expectancies (Year 3) as a mediator.

Results: Social media time (Year 2) was significantly associated with cannabis experimentation (Year 4) ($B = 0.18$, 95 % CI: 0.13–0.24, $p < 0.001$). At Year 3, positive cannabis expectancies predicted higher likelihood of experimentation one year later, while negative expectancies predicted lower likelihood. Positive cannabis expectancies mediated 19.8 % and negative cannabis expectancies mediated 13.6 % of the social media-cannabis use association.

Conclusions: Given the association between social media exposure on cannabis use and the mediating role of expectancies for adolescents, future research may explore interventions that limit both social media time and address expectancies to prevent early initiation of cannabis use.

1. Introduction

Cannabis is the third most widely used substance among adolescents in the United States (U.S.) after alcohol and tobacco, raising public health concerns about its impact on early adolescent health and

development (Substance Abuse and Mental Health Services Administration, 2025). Adolescent cannabis use has been linked to impairments in memory, executive function, and brain development, as well as with poorer school and lifetime achievement (Urits et al., 2021; Volkow et al., 2014). It has also been associated with long-term adverse health effects

* Correspondence to: 550 16th Street, 4th Floor, Box 0503, San Francisco, CA 94143, USA.

E-mail addresses: jason.nagata@ucsf.edu (J.M. Nagata), Andrew.Caffrey@ucsf.edu (A. Caffrey), nathan.nguyen2@ucsf.edu (N.D. Nguyen), snayak5@ucsf.edu (S. Nayak), isaac_frimpong@berkeley.edu (I. Frimpong), christiane@berkeley.edu (C.K. Helmer), colbey.ricklefs@ucsf.edu (C. Ricklefs), abubakr.alshoaibi@ucsf.edu (A.A. Al-Shoaibi), Alexander.Testa@uth.tmc.edu (A. Testa), Claire.Brindis@ucsf.edu (C.D. Brindis), Glenn-milo.Santos@ucsf.edu (G.-M. Santos), Fiona.Baker@sri.com (F.C. Baker).

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that extend into adulthood, including a greater risk of developing substance use disorder, psychiatric illness, cardiovascular disease, cerebrovascular disease, and peripheral disease (Hurd et al., 2019; Jackson et al., 2022; Volkow et al., 2014).

Social media stands out in its potential to shape cannabis expectancies, specifically given the widespread visibility of cannabis content and platform-driven engagement (Vogel et al., 2025). Expectancies are personal beliefs about the likely outcomes of a behavior (Torrealday et al., 2008). When it comes to cannabis experimentation, positive expectancies are beliefs that cannabis has positive effects, such as relaxation and tension reduction (Torrealday et al., 2008). In contrast, negative expectancies are beliefs that cannabis has negative effects, such as cognitive and behavioral impairment (Torrealday et al., 2008). Social learning theory may offer a framework for understanding how social media influences cannabis use behaviors (David, 2020; Smith, 2021). According to this theory, learned behaviors are facilitated through observation, including directly in real-world contexts or indirectly through media exposure (David, 2020). Digital media platforms such as Twitter/X, YouTube, Instagram, and TikTok commonly feature substance-related content, with over 76 % depicting substance use in a normalized or positive light (Rutherford et al., 2023). Exposure to such content can modulate adolescent beliefs and behaviors towards substance use and has been linked to greater intentions to use cannabis (Kelleghan et al., 2020; Kim et al., 2022). In addition, cannabis influencer marketing further amplifies this exposure by positioning influencers as credible and relatable sources, contributing to the normalization of substance use despite regulatory measures (Childers et al., 2019; Vassey et al., 2023).

During adolescence, youth are especially vulnerable to influence by the norms and approval of their peer groups (Hahlbeck and Vito, 2022). Social media amplifies this dynamic by providing constant exposure to not only peer behaviors and attitudes, but also substance-related content from brands, influencers, and celebrities. One study showed that over 34 % of adolescents reported exposure to substance-related content on social media at least weekly, and that exposure from different content sources was significantly associated with current e-cigarette and cannabis use (Vassey et al., 2025). Given this dynamic, it is crucial to address the implications of cannabis-related content exposure on social media.

Previous literature observes the connections between social media and health risk behaviors in adolescents, with one meta-analysis revealing significant associations between frequent social media use and increased engagement in gambling, alcohol, tobacco, and drug use (Purba et al., 2023). Past research has found that youth who endorsed more frequent social media use tended to have more favorable attitudes toward cannabis, which, in turn, were associated with higher odds of past-month cannabis use (Vogel et al., 2025). However, this study was state-specific and cross-sectional, and little research has prospectively examined social media use in relation to cannabis expectancies and subsequent cannabis use. The present study aimed to examine the prospective association between social media time and cannabis experimentation among a large, ethnically and geographically diverse, population-based sample of U.S. adolescents, with cannabis use expectancies as a mediator. We hypothesized that social media time would be associated with a higher risk of cannabis experimentation and that this relationship would be mediated by cannabis expectancies; greater social media use will lead to stronger positive expectancies, and subsequently higher risk for cannabis experimentation, whereas negative expectancies will reduce the risk for cannabis experimentation.

2. Methods

Prospective data were analyzed from the Adolescent Brain Cognitive Development (ABCD) Study. The ABCD Study is a longitudinal investigation of brain development and health in 11,875 children (ages 9–10 years at baseline, 2016–2018) recruited from 21 sites across the U.S.

Data were analyzed from follow-up Year 2 (2018–2020, social media time), Year 3 (2019–2021, cannabis expectancies), and Year 4 (2020–2022, cannabis experimentation). Participants with missing data for social media use, cannabis variables, or covariates were excluded, resulting in a total of 7691 participants in the analysis (Appendix A). Further descriptions of the study sample, recruitment, procedures, and measures have been described previously (Barch et al., 2018).

2.1. Independent variable: social media time

Self-reported social media time was determined from the ABCD Youth Screen Time Survey, based on a previously validated measure (Bagot et al., 2018; Paulus et al., 2019). Participants reported typical hours per day spent on social media (e.g., Facebook, Instagram, Twitter/X) separately for weekdays and weekend days. We performed a weighted average calculation of participants' typical weekday and weekend social media consumption to obtain a weekly average measure ($[\text{weekday average} \times 5] + [\text{weekend average} \times 2]/7$) (Nagata et al., 2022). We reported the weighted average social media time as a continuous variable measured in hours per day.

2.2. Dependent variable: cannabis experimentation

Cannabis experimentation was assessed using the Timeline Follow-Back (TLFB) instrument, a retrospective calendar-based self-report measure for substance use behaviors (Lisdahl et al., 2018; Sobell and Sobell, 1992). The TLFB has demonstrated strong psychometric properties in adolescent populations, including high test-retest reliability and convergent validity with other substance use measures (Fals-Stewart et al., 2000; Harris et al., 2016; Levy et al., 2004). We examined binary responses (yes/no) to TLFB items asking whether participants had experimented with marijuana or cannabis ("a puff or eaten any marijuana, also called pot, grass, weed or ganja?") since their last study visit, assessed at Year 4.

2.3. Mediating variable: cannabis expectancies

Cannabis expectancies were measured at Year 3 using the Marijuana Effect Expectancy Questionnaire-Brief (MEEQ-B), a 6-item instrument validated for adolescent populations (Torrealday et al., 2008). The MEEQ-B assesses beliefs about cannabis effects across six domains: cognitive and behavioral impairment, relaxation and tension reduction, social and sexual facilitation, perceptual and cognitive enhancement, global negative effects, and craving and physical effects. Participants rated each item on a 5-point Likert scale (1 = strongly disagree to 5 = strongly agree). The measure demonstrated adequate internal consistency in our sample (Cronbach's $\alpha = 0.60\text{--}0.77$) (Hussey et al., 2025; Taber, 2018) and has shown test-retest reliability in adolescent samples (Torrealday et al., 2008). Composite mean scores were calculated separately for positive and negative expectancies (Appendix B).

2.4. Covariates

We adjusted for age, sex (male or female), and race/ethnicity (Asian, Black, Latino/Hispanic, Native American, White, other), household income (six categories), highest parent education (high school or less vs college education or more), depressive symptoms via the Child Behavior Checklist t-scores (Achenbach and Rescorla, 2000), Year 2 cannabis experimentation (yes or no), friends' cannabis use (yes or no), and study site. Missing depressive symptoms data was imputed using the last observation carried forward method.

2.5. Statistical analysis

All analyses were conducted in Stata version 18 (StataCorp). Sequential modeling approaches were implemented, adjusting for

covariates and Year 2 cannabis experimentation. First, we examined the direct association between social media exposure and cannabis experimentation. Additionally, we evaluated the associations between social media exposure and positive and negative cannabis expectancies, as well as the association between cannabis expectancies and cannabis experimentation. To assess mediation, direct and indirect effects of cannabis expectancies on the social media-cannabis experimentation pathway were quantified using generalized structural equation modeling (GSEM). GSEM integrates aspects of path analysis, multiple regression, and confirmatory factor analysis to capture how sociodemographic and mediator data affect the outcome. GSEM then quantifies how the multifaceted variables influence the complex direct and indirect pathways leading to the study outcomes by considering the direct, indirect, and total effects of the variables used within this study (Endawkie et al., 2025). Assumptions for GSEM, which included binomial logit regressions for the binary outcomes, were evaluated and met. Multicollinearity was low (VIFs < 2), continuous predictors demonstrated linearity with the log-odds (Box-Tidwell $p > 0.05$), no influential outliers were detected (all Pregibon's $dbeta < 1$), observations were independent, the sample size was adequate, all outcomes were binary, and the models successfully converged. Sampling weights were applied to align ABCD Study demographics with American Community Survey estimates (Heeringa and Berglund, 2020). Statistical significance was set at $p < 0.05$ using two-tailed tests.

3. Results

Of the 7691 participants who met our inclusion criteria, 47.9 % were female with a mean age of 12.0 years ($SD=0.7$), and 42.6 % were racial/ethnic minorities (Table 1). At Year 4, 5.1 % reported having a puff of

Table 1
Sociodemographic and cannabis experimentation characteristics of 7691 Adolescent Brain Cognitive Development (ABCD) Study participants.

Sociodemographic and cannabis experimentation characteristics	Mean (SD) / (%)
Age (Year 2, years)	12.0 (0.7)
Sex (baseline)	
Female	47.9 %
Male	52.1 %
Race and ethnicity (baseline)	
Asian	5.2 %
Black	14.0 %
Latino/Hispanic	18.9 %
Native American	3.3 %
Other	1.3 %
White	57.4 %
Household income (Year 2)	
\$24,999 or less	13.1 %
\$25,000 to \$49,999	16.3 %
\$50,000 to \$74,999	16.9 %
\$75,000 to \$99,999	15.0 %
\$100,000 to \$199,999	28.1 %
\$200,000 or greater	10.6 %
Parent's highest education (Year 2)	
High school education or less	10.5 %
College education or more	89.5 %
CBCL depressive symptoms t-score (Year 2)	54.5 (6.3)
Social media, hours per day (Year 2)	0.7 (1.5)
Have had a puff/taste of cannabis since past visit	
Yes (Year 2)	0.2 %
Yes (Year 4)	5.1 %
Cannabis use by friends (Year 2)	
Yes	2.7 %
No	97.3 %
Cannabis expectancy score ^a (Year 3)	
Positive	7.8 (3.0)
Negative	11.9 (2.6)

Sampling weights were applied to yield representative estimates based on the American Community Survey from the US Census. SD = standard deviation, CBCL = Child Behavior Checklist. ^aCannabis expectancy score: possible and observed scoring range = 3–15.

cannabis in the prior year.

After adjusting for covariates, including friends' cannabis use, social media time at Year 2 was significantly associated with cannabis experimentation at Year 4 ($B=0.18$, 95 % CI: 0.13–0.24, $p < 0.001$) (Fig. 1). Social media time at Year 2 was associated with higher positive cannabis expectancies and lower negative cannabis expectancies one year later (Fig. 1). Positive cannabis expectancies at Year 3 were associated with higher likelihood of cannabis experimentation one year later, while negative cannabis expectancies at Year 3 were associated with lower likelihood of cannabis experimentation one year later (Fig. 1).

There were indirect effects of both positive expectancies ($B=0.05$, 95 % CI: 0.03–0.06, $p < 0.001$) and negative expectancies ($B=0.03$, 95 % CI: 0.02–0.04, $p < 0.001$) on the association between social media and cannabis experimentation. Positive cannabis expectancies mediated 19.8 % (95 % CI: 12.6 %–27.0 %) of the association, and negative cannabis expectancies mediated 13.6 % (95 % CI: 8.4 %–18.8 %) of the association (Fig. 1).

4. Discussion

The original hypothesis of this study states that social media time will be positively associated with cannabis experimentation among adolescents while being mediated by cannabis expectancies; i.e., positive expectancies lead to a higher risk for experimentation, while negative expectancies lower it. The analysis of social media time showed a statistically significant ($p < 0.001$) prospective association with a higher risk of cannabis experimentation one year later, even after accounting for friends' cannabis use. The association was partially mediated by positive and negative cannabis expectancies with statistically significant indirect effects, confirming our hypothesis. This aligns with prior findings (Vassey et al., 2025; Willoughby et al., 2022) while uniquely utilizing a larger and more demographically diverse cohort (Vassey et al., 2023; Willoughby et al., 2022).

Social learning theory provides a theoretical framework for the association between social media use and cannabis experimentation. According to this theory, adolescents learn through observing and modeling their own behaviors after others. Social media serves as a platform where adolescents are frequently exposed to cannabis-related content that normalizes use (Cavazos-Rehg et al., 2016; Kim et al., 2022). One study analyzing cannabis-related content on Twitter/X found that a significant majority (66 %) of tweets made by adolescents depict it in a positive context, with 43 % disclosing personal cannabis use online (Thompson et al., 2015). Beyond peer-driven content, exposure may also occur through sources such as brands, influencers, and celebrities, which may increase adolescents' positive expectancies about cannabis use (Vassey et al., 2023).

We build on earlier studies suggesting that frequent social media use in adolescents may foster pro-cannabis attitudes, which could increase the likelihood of future use (Vogel et al., 2025). Cannabis-related social media engagement and activity have also been associated with marijuana use in young people, independent of state-level policies and demographics (Kim et al., 2022). Exposure to social media posts about cannabis has been associated with a greater likelihood of cannabis use, including vaping, cannabis smoking, and using edible products (Vassey et al., 2025). Our findings confirm these prior associations, demonstrating that social media use is linked to greater positive cannabis expectancies and cannabis experimentation.

Additionally, we found that negative cannabis expectancies were associated with lower levels of cannabis experimentation. Adolescents who used social media less frequently reported greater negative expectancies for cannabis, suggesting that both lower social media exposure and stronger negative beliefs about cannabis may serve as protective factors against cannabis experimentation. The present study contributes to previous findings by specifically focusing on an early adolescent population and identifying both positive and negative expectancies as partial mediating mechanisms (Vogel et al., 2025).

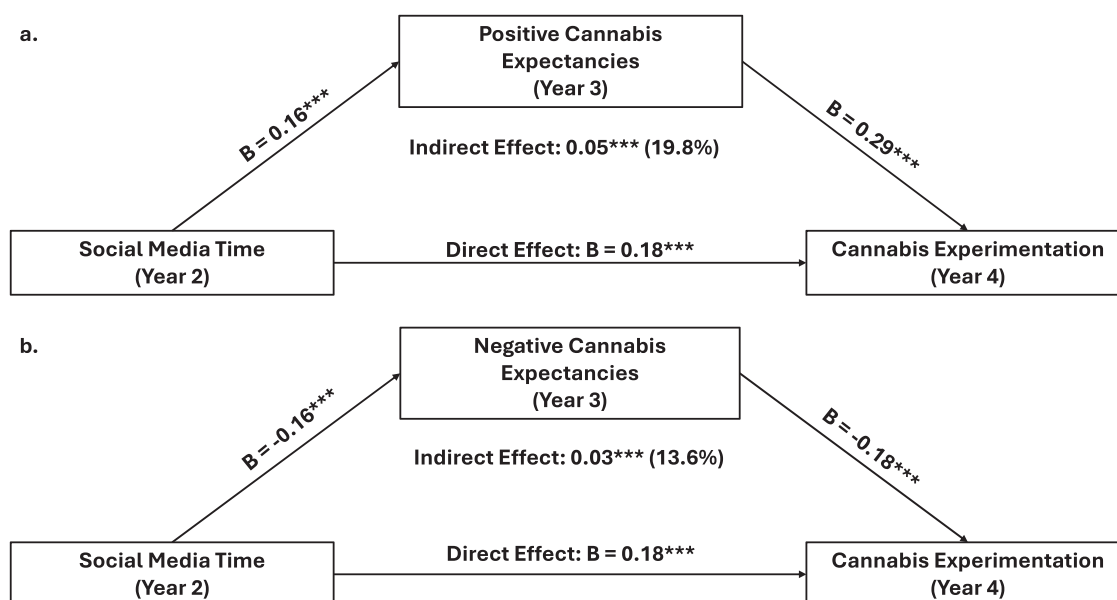


Fig. 1. Direct and indirect (via cannabis expectancies) effects of social media on cannabis experimentation. Asterisks denote statistical significance: (***) for $p < 0.001$. a) Mediation of positive cannabis expectancies. b) Mediation of negative cannabis expectancies. Models adjusted for age, sex, race/ethnicity, household income, parent education, depressive symptoms, Year 2 cannabis experimentation, friends' cannabis use, and study site.

Limitations include the usage of self-reported data, which may introduce potential biases. Social media was assessed without accounting for variations across platforms and did not consider distinctions between the nature of social media engagement, such as passive viewing versus active viewing. Furthermore, the absence of data on specific social media content prevented us from estimating participants' true exposure to cannabis-related content. Findings partially mediated by both positive and negative expectancies may indicate an incomplete explanation of the mechanism. Cannabis-related perceptions, adolescent personality traits, and trauma history are all potential mechanisms that may influence cannabis use and should be explored more thoroughly in future research (Brammer et al., 2022; Pilatti et al., 2021; Pilin et al., 2022).

One of the strengths of the present study is its prospective design, which captured social media, cannabis expectancies, and cannabis experimentation in the appropriate temporal sequence. The study's focus is on early adolescents, a key adolescent demographic for early preventive intervention efforts. Additionally, the use of a demographically diverse sample enhanced the relevance and generalizability of the findings. Lastly, the observed association between social media use and later cannabis experimentation remained significant even after adjusting for friends' cannabis use, suggesting a robust and independent effect of social media exposure.

5. Conclusion

Developing a deeper understanding of the determinants of cannabis use in adolescence is crucial in formulating early prevention frameworks and guiding public health policy development. Policymakers should impose stricter regulations by enacting policies that hold platforms accountable for allowing substance-related content to reach an early adolescent audience (Fassi et al., 2024). Policies aimed at mitigating early adolescent exposure to cannabis-related content on social media—such as strengthened parent controls, robust age verification, and enhanced privacy controls—should be prioritized (Bozzola et al., 2022; Radesky et al., 2020). Clinicians can incorporate guidance by addressing potential harms of high-risk social media platform usage in situations where adolescents may be most susceptible to substance use (Fassi et al., 2024). Future studies should examine platform- and

content-specific social media use and their associations with cannabis-related outcomes in adolescents.

CRediT authorship contribution statement

Jason M. Nagata: Writing – review & editing, Writing – original draft, Supervision, Formal analysis, Conceptualization. **Andrew Caffrey:** Writing – review & editing, Writing – original draft, Formal analysis. **Nathan D. Nguyen:** Writing – review & editing, Writing – original draft. **Sahana Nayak:** Writing – review & editing, Writing – original draft. **Isaac Frimpong:** Writing – review & editing, Writing – original draft. **Christiane K. Helmer:** Writing – review & editing, Writing – original draft, Formal analysis. **Colbey Ricklefs:** Writing – review & editing, Writing – original draft. **Abubakr A. Al-Shoaibi:** Writing – review & editing, Writing – original draft, Formal analysis. **Alexander Testa:** Conceptualization. **Claire D. Brindis:** Writing – review & editing. **Glenn-Milo Santos:** Writing – review & editing. **Fiona C. Baker:** Writing – review & editing, Data curation, Conceptualization.

Ethics approval

Centralized institutional review board (IRB) approval was obtained from the University of California, San Diego. Study sites obtained approval from their respective IRBs, caregivers provided written informed consent, and each child provided written assent.

Consent to participate

Not applicable

Consent to publish

Not applicable

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Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Appendix A. Supporting information

Supplementary data associated with this article can be found in the online version at [doi:10.1016/j.drugalcdep.2025.112947](https://doi.org/10.1016/j.drugalcdep.2025.112947).

Data Availability

Data used in the preparation of this article were obtained from the ABCD Study (<https://abcdstudy.org>), held in the NIH Brain Development Cohorts Portal.

References

- Achenbach, T.M., Rescorla, L.A., 2000. Manual for the ASEBA preschool forms and profiles. Research Center for Children, Youth, & Families, Burlington, VT.
- Bagot, K.S., Matthews, S.A., Mason, M., Squeglia, L.M., Fowler, J., Gray, K., Herting, M., May, A., Colrain, I., Godino, J., Tapert, S., Brown, S., Patrick, K., 2018. Current, future and potential use of mobile and wearable technologies and social media data in the ABCD Study to increase understanding of contributors to child health. *Dev. Cogn. Neurosci.* 32, 121–129. <https://doi.org/10.1016/j.dcn.2018.03.008>.
- Barch, D.M., Albaugh, M.D., Avenevoli, S., Chang, L., Clark, D.B., Glantz, M.D., Hudziak, J.J., Jernigan, T.L., Tapert, S.F., Yurgelun-Todd, D., Alia-Klein, N., Potter, A.S., Paulus, M.P., Prouty, D., Zucker, R.A., Sher, K.J., 2018. Demographic, physical and mental health assessments in the Adolescent Brain and Cognitive Development Study: rationale and description. *Dev. Cogn. Neurosci.* 32, 55–66. <https://doi.org/10.1016/j.dcn.2017.10.010>.
- Bozzola, E., Spina, G., Agostiniani, R., Barni, S., Russo, R., Scarpatto, E., Di Mauro, A., Di Stefano, A.V., Caruso, C., Corsello, G., Staiano, A., 2022. The use of social media in children and adolescents: scoping review on the potential risks. *Int. J. Environ. Res. Public Health* 19, 9960. <https://doi.org/10.3390/ijerph19169960>.
- Brammer, W.A., Conn, B.M., Iverson, E., Lankenau, S.E., Dodson, C., Wong, C.F., 2022. Coping motives mediate the association of trauma history with problematic cannabis use in young adult medical cannabis patients and non-patient cannabis users. *Subst. Use Misuse* 57, 684–697. <https://doi.org/10.1080/10826084.2022.2026970>.
- Cavazos-Rehg, P.A., Krauss, M.J., Sowles, S.J., Bierut, L.J., 2016. Marijuana-related posts on Instagram. *Prev. Sci.* 17, 710–720. <https://doi.org/10.1007/s11212-016-0669-9>.
- Childers, C.C., Lemon, L.L., Hoy, M.G., 2019. Sponsored #Ad: agency perspective on influencer marketing campaigns. *J. Curr. Issues Res. Advert.* 40, 258–274. <https://doi.org/10.1080/10641734.2018.1521113>.
- David, L., 2020. Social Learning Theory (Bandura) [WWW Document]. Learning Theories. URL (<https://learning-theories.com/social-learning-theory-bandura.html>).
- Endawkie, A., Keleb, A., Dilnesa, T., Tsenga, Y., 2025. Generalized structural equation modeling of direct and indirect determinants of chronic undernutrition among under-five children in Ethiopia: further analysis of the 2019 mini Ethiopian demographic and health survey. *J. Health Popul. Nutr.* 44, 73. <https://doi.org/10.1186/s41043-025-00792-8>.
- Fals-Stewart, W., O'Farrell, T.J., Freitas, T.T., McFarlin, S.K., Rutigliano, P., 2000. The Timeline Followback reports of psychoactive substance use by drug-abusing patients: psychometric properties. *J. Consult. Clin. Psychol.* 68, 134–144. <https://doi.org/10.1037/0022-006x.68.1.134>.
- Fassi, L., Thomas, K., Parry, D.A., Leyland-Craggs, A., Ford, T.J., Orben, A., 2024. Social media use and internalizing symptoms in clinical and community adolescent samples: a systematic review and meta-analysis. *JAMA Pediatr.* 178, 814–822. <https://doi.org/10.1001/jamapediatrics.2024.2078>.
- Hahlbeck, S.M., Vito, A.G., 2022. Adolescent marijuana dependence: the role of social bonds and social learning theory. *J. Psychoact. Drugs* 54, 43–53. <https://doi.org/10.1080/02791072.2021.1903122>.
- Harris, S.K., Knight, J.R., Van Hook, S., Sherritt, L., Brooks, T.L., Kulig, J.W., Nordt, C.A., Saitz, R., 2016. Adolescent substance use screening in primary care: validity of computer self-administered versus clinician-administered screening. *Subst. Abuse* 37, 197–203. <https://doi.org/10.1080/08897077.2015.1014615>.
- Heeringa, S., Berglund, P., 2020. A guide for population-based analysis of the Adolescent Brain Cognitive Development (ABCD) Study baseline data, 02.10.942011. *bioRxiv* 2020. <https://doi.org/10.1101/2020.02.10.942011>.
- Hurd, Y.L., Manzoni, O.J., Pletnikov, M.V., Lee, F.S., Bhattacharyya, S., Melis, M., 2019. Cannabis and the developing brain: insights into its long-lasting effects. *J. Neurosci.* 39, 8250–8258. <https://doi.org/10.1523/JNEUROSCI.1165-19.2019>.
- Hussey, I., Alsalti, T., Bosco, F., Elson, M., Arslan, R., 2025. An aberrant abundance of Cronbach's Alpha Values at.70, 25152459241287123. *Adv. Methods Pract. Psychol. Sci.* 8. <https://doi.org/10.1177/25152459241287123>.
- Jackson, D.B., Testa, A., Woodward, K.P., Qureshi, F., Ganson, K.T., Nagata, J.M., 2022. Adverse childhood experiences and cardiovascular risk among young adults: findings from the 2019 behavioral risk factor surveillance system. *Int. J. Environ. Res. Public Health* 19, 11710. <https://doi.org/10.3390/ijerph191811710>.
- Kelleghan, A.R., Leventhal, A.M., Cruz, T.B., Bello, M.S., Liu, F., Unger, J.B., Riehm, K., Cho, J., Kirkpatrick, M.G., McConnell, R.S., Barrington-Trimis, J.L., 2020. Digital media use and subsequent cannabis and tobacco product use initiation among adolescents. *Drug Alcohol Depend.* 212, 108017. <https://doi.org/10.1016/j.drugalcdep.2020.108017>.
- Kim, Y., Vera, L., Huang, J., Emery, S., 2022. Marijuana content on digital media and marijuana use among young people in the United States. *Cannabis* 5, 74–84. <https://doi.org/10.26828/cannabis/2022.02.007>.
- Levy, S., Sherritt, L., Harris, S.K., Gates, E.C., Holder, D.W., Kulig, J.W., Knight, J.R., 2004. Test-retest reliability of adolescents' self-report of substance use. *Alcohol Clin. Exp. Res.* 28, 1236–1241. <https://doi.org/10.1097/01.alc.0000134216.22162.a5>.
- Lisdahl, K.M., Sher, K.J., Conway, K.P., Gonzalez, R., Feldstein Ewing, S.W., Nixon, S.J., Tapert, S., Bartsch, H., Goldstein, R.Z., Heitzeg, M., 2018. Adolescent Brain Cognitive Development (ABCD) Study: overview of substance use assessment methods. *Dev. Cogn. Neurosci.* 32, 80–96. <https://doi.org/10.1016/j.dcn.2018.02.007>.
- Nagata, J.M., Ganson, K.T., Iyer, P., Chu, J., Baker, F.C., Pettee Gabriel, K., Garber, A.K., Murray, S.B., Bibbins-Domingo, K., 2022. Sociodemographic correlates of contemporary screen time use among 9- and 10-year-old children. *J. Pediatr.* 240, 213–220.e2. <https://doi.org/10.1016/j.jpeds.2021.08.077>.
- Paulus, M.P., Squeglia, L.M., Bagot, K., Jacobus, J., Kuplicki, R., Breslin, F.J., Bodurka, J., Morris, A.S., Thompson, W.K., Bartsch, H., Tapert, S.F., 2019. Screen media activity and brain structure in youth: evidence for diverse structural correlation networks from the ABCD study. *NeuroImage* 185, 140–153. <https://doi.org/10.1016/j.neuroimage.2018.10.040>.
- Pilatti, A., Prince, M.A., Bravo, A.J., Pearson, M.R., Mezquita, L., Pautassi, R.M., Cross-Cultural Addictions Study Team, 2021. Cannabis-related perceptions as mediators of the association between trait impulsivity and cannabis outcomes. *J. Stud. Alcohol Drugs* 82, 522–535. <https://doi.org/10.15288/jsad.2021.82.522>.
- Pilin, M.A., Robinson, J.M., Young, K., Krank, M.D., 2022. Cognitions mediate the influence of personality on adolescent cannabis use initiation. *Addict. Behav. Rep.* 15, 100425. <https://doi.org/10.1016/j.abrep.2022.100425>.
- Purba, A.K., Thomson, R.M., Henery, P.M., Pearce, A., Henderson, M., Katikireddi, S.V., 2023. Social media use and health risk behaviours in young people: systematic review and meta-analysis. *BMJ* 383, e073552. <https://doi.org/10.1136/bmj-2022-073552>.
- Radesky, J., Chassiakos, Y.R., Ameenuddin, N., Navsaria, D., Ameenuddin, N., Boyd, R., Selkie, E., Radesky, J., Patrick, M., Friedman, J., Evans, Y., Tomopoulos, S., Bhargava, H., Hutchinson, J., Bracho-Sanchez, E., 2020. Digital advertising to children. *Pediatrics* 146, e20201681. <https://doi.org/10.1542/peds.2020-1681>.
- Rutherford, B.N., Lim, C.C.W., Johnson, B., Cheng, B., Chung, J., Huang, S., Sun, T., Leung, J., Stjepanović, D., Chan, G.C.K., 2023. TurnTrending: a systematic review of substance use portrayals on social media platforms. *Addiction* 118, 206–217. <https://doi.org/10.1111/add.16020>.
- Smith, M.A., 2021. Social learning and addiction. *Behav. Brain Res.* 398, 112954. <https://doi.org/10.1016/j.bbr.2020.112954>.
- Sobell, L.C., Sobell, M.B., 1992. Timeline follow-back: A technique for assessing self-reported alcohol consumption, in: *Measuring Alcohol Consumption: Psychosocial and Biochemical Methods*. Humana Press/Springer Nature, Totowa, NJ, US, pp. 41–72. https://doi.org/10.1007/978-1-4612-0357-5_3.
- Substance Abuse and Mental Health Services Administration, 2025. Key substance use and mental health indicators in the United States: results from the 2024 National Survey on Drug Use and Health (HHS Publication No. PEP25-07-007, NSDUH Series H-60). Center for Behavioral Health Statistics and Quality, Substance Abuse and Mental Health Services Administration.
- Taber, K.S., 2018. The use of Cronbach's Alpha when developing and reporting research instruments in science education. *Res. Sci. Educ.* 48, 1273–1296. <https://doi.org/10.1007/s11165-016-9602-2>.

- Thompson, L., Rivara, F.P., Whitehill, J.M., 2015. Prevalence of marijuana-related traffic on Twitter, 2012-2013: a content analysis. *Cyber Behav. Soc. Netw.* 18, 311–319. <https://doi.org/10.1089/cyber.2014.0620>.
- Torrealdy, O., Stein, L. a R., Barnett, N., Golembeske, C., Lebeau, R., Colby, S.M., Monti, P.M., 2008. Validation of the Marijuana effect expectancy questionnaire-brief. *J. Child Adolesc. Subst. Abus.* 17, 1. <https://doi.org/10.1080/15470650802231861>.
- Urits, I., Charipova, K., Gress, K., Li, N., Berger, A.A., Cornett, E.M., Kassem, H., Ngo, A. L., Kaye, A.D., Viswanath, O., 2021. Adverse effects of recreational and medical cannabis. *Psychopharmacol. Bull.* 51, 94–109.
- Vassey, J., Cho, J., Vogel, E.A., Iyer, T., Chen-Sankey, J., Unger, J.B., 2025. E-cigarette and cannabis social media posts and adolescent substance use. *JAMA Netw. Open* 8, e2517611. <https://doi.org/10.1001/jamanetworkopen.2025.17611>.
- Vassey, J., Valente, T., Barker, J., Stanton, C., Li, D., Laestadius, L., Cruz, T.B., Unger, J. B., 2023. E-cigarette brands and social media influencers on Instagram: a social network analysis. *Tob. Control* 32, e184–e191. <https://doi.org/10.1136/tobaccocontrol-2021-057053>.
- Vogel, E.A., Romm, K.F., Massey, Z.B., Cohn, A.M., 2025. Social media use, cannabis attitudes, and cannabis use among adolescents: an observational study. *Prev. Med. Rep.* 55, 103126. <https://doi.org/10.1016/j.pmedr.2025.103126>.
- Volkow, N.D., Baler, R.D., Compton, W.M., Weiss, S.R.B., 2014. Adverse health effects of Marijuana use. *N. Engl. J. Med.* 370, 2219–2227. <https://doi.org/10.1056/NEJMr1402309>.
- Willoughby, J.F., Hust, S.J.T., Li, J., Couto, L., 2022. Social media, Marijuana and sex: an exploratory study of adolescents' intentions to use and college students' use of Marijuana. *J. Sex. Res.* 59, 85–97. <https://doi.org/10.1080/00224499.2020.1827217>.